

BEYOND RPA • ADVANCE AGENTIC AUTOMATION

Advance Agentic RPA & UiPath Architecture

A 35-session masterclass covering Document Intelligence, AI Center ML pipelines, Studio Autopilot, Self-Healing Selectors, Model Context Protocol (MCP) Servers, Python Coded Agents, and UiPath Maestro Multi-Agent Swarms.

THE PARADIGM SHIFT: BEYOND TRADITIONAL RPA

Transition from brittle screen scraping and fixed flowchart bots to autonomous reasoning agents equipped with dynamic tool registries, validation guardrails, and enterprise BPMN orchestration.

SESSIONS

35

Live interactive builds

MODULES

12

Enterprise architecture

PHASES

3

Structured progression

FORMAT

100%

Hands-on code repos

[Explore Course Webpage →](#)[Talk to RPA Vault Mentors →](#)

CURRICULUM ARCHITECTURE

Beyond RPA: *Architecture for the Modern AI Stack*

Stop training for the automation platform of two years ago. Master the full stack of intelligent document understanding, reasoning agents, and MCP protocols.

SESSIONS

35 Live

Code builds

35

MODULES

12 Deep

Full stack

12

PHASES

3 Stages

Zero to agentic

03

BUILDS

100% Code

Git repos

100%

THE PAST

01

Where Most Courses Stop

Legacy RPA methodologies causing production maintenance nightmares:

- **Brittle Selectors:** Flows break on DOM updates.
- **Hardcoded Regex:** Inflexible rules on variable layouts.
- **Manual Try/Catch:** Linear error traps without LLMs.
- **Siloed Bots:** Incapable of multi-agent swarm flows.

THE FUTURE

02

What You Build in This Track

Production-grade Agentic systems built for modern enterprise scale:

- **Autonomous LLM Agents:** Grounding & system prompts.
- **Model Context Protocol:** Tool APIs for LLMs.
- **Self-Healing Selectors:** ScreenPlay vision runtime.
- **Maestro BPMN Swarms:** Multi-agent coordination.

AUDIENCE

03

Who This Is Designed For

Tailored specifically for engineers seeking senior career acceleration:

- **UiPath Developers:** Moving past screen scraping.
- **RPA Architects:** Designing enterprise AI stacks.
- **Senior Engineers:** Targeting ₹12-20L+ packages.

PREREQUISITES

04

What You Need & Prerequisites

Zero AI/ML background needed — all concepts taught from first principles:

- **Studio Basics:** Familiarity with variables & flows.
- **Zero ML Required:** AI Center taught from scratch.
- **Python FastMCP:** Practical scripting taught step-by-step.

THE AGENTIC ADVANTAGE

Build systems that reason, heal, and orchestrate across the modern AI stack.

Learn Document Intelligence · Deploy AI Center ML models · Build Python MCP Servers · Orchestrate Multi-Agent Swarms with Maestro.

PHASE 01 • CORE FOUNDATIONS

4-Stage Document Understanding *Framework*

Master document digitization, intelligent taxonomy schemas, and production invoice extraction pipelines.

Day 01 DU Fundamentals & Enterprise Document Types

Structured, semi-structured, and unstructured document ingestion. OCR digitizer engines comparison (OmniPage, Google Cloud OCR, Microsoft OCR) and enterprise DU ROI.

Day 02 The 4-Stage DU Framework & Taxonomy Manager

Deep dive into Taxonomy Manager schema definitions, document type hierarchies, Digitization OCR engines, Keyword/Intelligent Classifiers, and Extractor orchestration.

Day 03 End-to-End Production Invoice Processing Capstone **CAPSTONE BUILD**

Full practical build implementing automated invoice processing, line-item table extractions, regex rule validation, exception routing, and enterprise debugging.

Day 04 Studio Document Understanding Framework Template

Understanding and customizing the enterprise Document Understanding framework template inside UiPath Studio, config asset integration, and state machine transitions.

Day 05 Modern DU Architecture & Pre-Trained Models

Comparing traditional DU vs Modern DU, utilizing out-of-the-box pre-trained document models (Invoices, Receipts, Purchase Orders), and establishing new project pipelines.

01 Taxonomy

Schema mapping & field grouping JSON.

02 Digitization

Multi-engine OCR text & DOM parsing.

03 Classification

Keyword & Intelligent vector routing.

04 Extraction

Pre-trained ML & line-item parsing.

PRACTICE LAB & CHECKPOINT: Build a complete Document Understanding pipeline with taxonomy schema definition, OCR digitizer calibration, and fallback regex extraction rules.

Have questions about Document Understanding? Connect with our mentors

[Schedule Mentor Call →](#)

PHASE 01 • ENTERPRISE MODULES

AI Center ML Retraining, *Action Center & IXP*

Deploy ML models, orchestrate human-in-the-loop validation tasks, and extract multi-page unstructured layouts.

DAY 06

06

AI Center ML Training & Pipelines

Deploying ML packages, building dataset pipelines, retraining models with enterprise domain documents, and consuming ML Skills in Studio.

- ML Packages & Dataset Pipelines
- Retraining with Domain Documents
- ML Skill Activity Deployment

DAY 07

07

Action Center & Human-in-the-Loop

Human-in-the-Loop validation concepts, creating validation tasks, Action Center user routing workflows, and approval orchestration.

- Creating HITL Validation Tasks
- Action Center Orchestrator Catalogs
- User Routing & Multi-Tier Approvals

DAY 08

08

Validation Station & Thresholds

Presenting Validation Station to users, configuring confidence thresholds (<85%), field-level overrides, and validation station best practices.

- Confidence Threshold Rules (<85%)
- Field-Level Exception Overrides
- Validation Station Best Practices

DAY 09

09

REST API Workflows & Web Services

REST API architecture, executing HTTP Request activities, parsing JSON responses, and integrating web services into Document Understanding.

- HTTP Request & OAuth 2.0 Auth
- JSON Deserialization & Data Tables
- ERP Ingestion Web Services

DAY 10

10

Intelligent Xtraction (IXP) Architecture

What Intelligent Xtraction & Processing (IXP) is, why it is required for complex layouts, and IXP vs traditional extraction comparison.

- IXP Next-Gen Extraction Engine
- Complex Unstructured Layouts
- IXP vs Traditional DU Comparison

DAY 11

11

Live IXP Multi-Page Capstone

Real-time production use case implementing IXP on complex unstructured multi-page documents with end-to-end downstream ERP sync.

- Multi-Page PDF Processing
- Complex Table Reconciliation
- End-to-End Downstream Sync

PRACTICE LAB & EXIT CHECK: Build a complete Document Understanding pipeline with AI Center ML skill integration, Action Center validation triggers on low confidence (<85%), and downstream REST API ERP sync.

Want to see a live demo of AI Center & IXP pipelines?

[Request Demo Access →](#)

PHASE 02 • ADVANCED AI

Autopilot, Self-Healing Selectors & Vision

Master Generative AI activities, Autopilot in Studio, Clipboard AI, Self-Healing Agents, ScreenPlay, and Semantic Activities.

DAYS 12-14

12-14

Studio Gen AI Activities, Autopilot & Clipboard AI

Generating workflows from plain English, controlling LLM parameters, and automating cross-application data flow:

Day 12

Gen AI Activities in Studio

Prompt grounding, temperature tuning, system instructions, and real-time business summarization.

Day 13

Autopilot in Studio & Workflow Synthesis

Generating complex automation workflows from natural language prompts and automated expression building.

Day 14

Clipboard AI Cross-App Transfer

Cross-app intelligent copy-paste and autonomous field mapping between spreadsheets, web portals, and legacy desktop apps.

Gen AI Activities

Autopilot Studio

Clipboard AI

Prompt Grounding

DAYS 15-17

15-17

Self-Healing Agents, ScreenPlay Vision & Semantic Actions

Building resilient bots that automatically repair broken selectors, see remote desktops with computer vision, and execute business intent:

Day 15

Self-Healing Agents & Dynamic Recovery

Dynamic selector recovery, self-healing runtime execution, and zero maintenance when web interfaces change.

Day 16

ScreenPlay Vision-Based Automation

Next-generation vision-based automation patterns for automating challenging remote desktop environments.

Day 17

Semantic Activities & Intent Execution

Executing automation tasks using natural language business intent rather than brittle CSS or XPath selectors.

Self-Healing Runtime

ScreenPlay Vision

Semantic Intent

Remote Desktop AI

Dynamic UI Healing

Fuzzy matching & visual anchor backup recovery.

Autopilot Studio

Natural language activity & LINQ synthesis.

Semantic Execution

Intent-driven actions over brittle DOM selectors.

PRACTICE LAB & EXIT CHECK: Deploy a self-healing bot utilizing ScreenPlay vision that automatically recovers from simulated UI element updates and passes semantic parameters to Gen AI activities.

Upgrade your UiPath skillset with Agentic AI and ScreenPlay Vision

Apply on Course Webpage →

PHASE 03 • AUTONOMOUS AGENTS

Agent Builder, Guardrails & AgentScore Matrix

Build reasoning agents in UiPath Agent Builder, define system prompt guardrails, and evaluate agent performance with Agent Score.

DAYS 18-21

18-21

Agent Builder & Prompt Guardrails Architecture

Transitioning from deterministic automation to autonomous decision loops with structured system instructions and memory context:

Day 18

Agentic AI Fundamentals & Agent Builder

Agent vs traditional automation architecture, autonomous decision loops, and the Agent Builder platform.

Day 19

Agent Builder Core Concepts & Memory

Context grounding, system instructions, short/long-term memory, and tool integration.

Day 20

Prompt Engineering & Guardrails

Techniques for prompt engineering, system instructions, few-shot prompting, schema validation, and guardrails.

Day 21

Build Your First Autonomous Agent

Live production build configuring autonomous agent parameters, input/output schemas, and testing.

Agent Builder

System Instructions

Few-Shot Prompting

Guardrails

DAYS 22-25

22-25

HITL Escalations & AgentScore Benchmarking

Governing agent behavior in production: measuring reasoning accuracy, stress-testing edge cases, and invoking agents from RPA:

Day 22

Agent Escalations (HITL) & Action Center

Human validation, escalation thresholds, approval workflows, and Action Center integration.

Day 23

Agent Evaluations & Agent Score

Evaluation benchmarks, metrics, ground truth datasets, and Agent Score calculation.

Day 24

Agent Tools Selection & Edge Cases

Evaluating agent accuracy, testing edge cases, tool selection benchmarks, and continuous monitoring.

Day 25

Bidirectional Agent + RPA Integration

Invoking autonomous agents from deterministic RPA workflows and passing structured context bidirectionally.

Agent Score

Evaluation Datasets

Action Center Escalation

RPA Invocation

System Guardrails

Strict schema validation & hallucination guards.

AgentScore Matrix

Ground-truth 50-item evaluation benchmarks.

HITL Routing

Action Center escalation on low confidence.

PRACTICE LAB & EXIT CHECK: Create an autonomous Customer Escalation Agent with custom prompt guardrails, measure its Agent Score over a 50-item evaluation dataset, and trigger Action Center tasks on uncertainty.

Learn how to build production-grade Agent Builder systems

Talk to an RPA Architect →

RPA Vault • Advance Agentic UiPath Curriculum

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PHASE 04 • FLAGSHIP LAYER

MCP Servers, Python Coded Agents & *Maestro Swarms*

Build Model Context Protocol (MCP) servers in Python, code custom agents, and orchestrate multi-agent swarms with UiPath Maestro.

MODULE 01 • DAYS 26-30

26-30

The Model Context Protocol (MCP) Server Suite

Mastering Anthropic's open standard for connecting LLMs to enterprise tools across UiPath, Python, and local CLIs.

Day 26 Introduction to Model Context Protocol (MCP)

What Model Context Protocol is, why it matters for autonomous agents, and client-server tool registration architecture.

Day 27 UiPath MCP Server Configuration & Tool Registration

Practical demo configuring and registering enterprise automation tools inside the native UiPath MCP Server.

Day 28 Coded MCP Server Development with Python + UiPath

Practical demo building a custom Coded MCP Server using Python + UiPath to expose custom business logic to LLMs.

Day 29 Command MCP Server for Local CLI & Process Tools

Local tool execution, process invocation, and command-line Model Context Protocol integration.

Day 30 Remote MCP Server Enterprise Deployment & Security

Distributed enterprise MCP server deployment, remote endpoint hosting, and network security protocols.

MODULE 02 • DAYS 31-32

31-32

Python Coded Agents & AI Coding

Writing pure Python automation scripts, deploying to Orchestrator, and synthesizing workflows using Claude 3.5 Sonnet & Codex.

Day 31 Python Coded Agents

Building a Coded Agent in Python and deploying to Orchestrator.

Day 32 AI Coding Skills (Claude & Codex)

Synthesizing and optimizing RPA workflows automatically.

MODULE 03 • DAYS 33-35

33-35

UiPath Maestro & CrewAI Swarms

BPMN 2.0 process modeling, state management, event orchestration, and coordinating autonomous CrewAI swarms.

Day 33 Maestro BPMN 2.0 Flows

BPMN standard, tasks, events, and decision gateways.

Days 34-35 Multi-Agent Swarm Demo

Executing swarms with external CrewAI frameworks.

FLAGSHIP CAPSTONE ARCHITECTURE PROJECT: Architect an end-to-end autonomous enterprise system connecting a Python FastMCP Server, an Agent Builder reasoning agent, and a UiPath Maestro BPMN gateway for end-to-end order fulfillment.

The flagship layer separating you from the market

[View Full Capstone Details →](#)

WHAT'S INCLUDED • LIVE DELIVERY

Everything you need to master *next-generation automation*

Live interactive coding, production codebases, 1-on-1 mentor code reviews, and industry-recognized credentials.

PILLAR 01

01

Live Instructor-Led Builds

Interactive coding sessions in every class — not pre-recorded slides or generic videos. You build real pipelines alongside senior architects.

- Live step-by-step code walkthroughs
- Real-time debugging & problem solving

PILLAR 02

02

Production-Grade Architecture

Build real Document Understanding pipelines, AI Center ML models, Python MCP servers, Coded Agents, and Maestro BPMN flows.

- Enterprise design patterns & standards
- Production error handling & guardrails

PILLAR 03

03

Lifetime Recording & Code Access

Full HD recordings of every class with complete project source code repositories for continuous reference and production reuse.

- Full HD video recordings & timestamps
- Clean GitHub repositories for all modules

PILLAR 04

04

Direct Mentor Code Review

1-on-1 architecture guidance and pull-request reviews on your agent workflows from practicing RPA and AI architects.

- Personalized 1-on-1 feedback sessions
- Direct Git pull-request code reviews

PILLAR 05

05

Private Practitioner Community

Exclusive network of working RPA architects and Agentic AI developers sharing live job openings, project tips, and interview insights.

- Verified job leads & interview prep
- Architect networking & knowledge base

PILLAR 06

06

Industry-Recognized Credential

Official Certificate of Completion recognized across top enterprise employers, consulting firms, and hiring partners.

- Verifiable digital certificate ID
- Showcase on LinkedIn & resume

READY TO CLOSE THE GAP?

Advance your career to the forefront of Agentic Automation.

35 sessions. Three phases. One destination: agentic. Connect with RPA Vault mentors to enroll in the next live batch.

Limited batch sizes for personalized mentor code reviews

[Schedule Admissions Call →](#)

YOUR NEXT PRACTICAL STEP

Learn the tools. Build the evidence.

Explore the complete course, review the 35-session curriculum, and connect with RPA Vault mentors to plan your advanced automation journey.

Explore the complete course

See the full curriculum, learning model, project path, and course details online.

rpavault.com/course/advance-agentic-rpa-uipath/ →

Talk to RPA Vault Mentors

Discuss your background, course fit, batch options, and enrollment steps.

rpavault.com/contact/ →

[Explore Course Webpage →](https://rpavault.com/course/advance-agentic-rpa-uipath/)[Talk to RPA Vault Mentors →](https://rpavault.com/contact/)